Proposal for Research

Clean Water Access in the Mexican State of Morelos

Abstract

Clean water access continues to be a critical challenge for over 2.2 billion people worldwide. In Mexico, where 57% of households lack safe water, addressing disparities in water access is necessary to improve public health within vulnerable communities. This research project aims to explore the relationships between population data and water qualities in the state of Morelos, Mexico. By analyzing data from Mexico's National Water Quality Measurement Network (RENAMECA) and population density data from the National institute of Statistics and Geography (INEGI), this study will investigate whether more densely populated areas have greater access to clean water. Additionally, communities without proper access to water will be identified. ArcGIS will be utilized to map out the water quality data and proximity to communities. The study findings will bring valuable insights for organizations, guiding efforts to improve data on water access in underserved communities. The project's broader goal is to contribute to efforts aimed at reducing water inequalities in Mexico, ultimately aiming towards better health outcomes and equitable access to resources.

Introduction and Background

As global population levels continue to rise, access to clean water remains a critical issue for human health. According to the World Health Organization, there are an estimated 2.2 billion individuals who do not have access to clean water. Without clean water, populations are unable to maintain proper sanitation. This can lead to the spread of waterborne diseases such as cholera, diarrhea, dysentery, hepatitis A, typhoid, and polio (World Health Organization). These diseases disproportionately affect vulnerable communities, which often lack access to modernized transportation infrastructure, making it difficult to reach lifesaving healthcare. Many of these vulnerable populations are found in developing countries, such as Mexico.

Mexico, a country with a population of over 130 million, continues to struggle with wealth disparity, leaving 36.3% of its population still living in poverty (BBVA). According to Planet Water Foundation, 57% of households in Mexico lack access to a safe water supply (Planet Water Foundation). Even in densely populated areas with more advanced infrastructure, clean water access may still be limited, furthering this issue. This project aims to investigate whether towns and municipalities within Mexican states such as Morelos have proper access to clean water.

I hypothesize that more densely populated areas will have greater access to clean water compared to less populated regions. To test this hypothesis, I will analyze pre-collected survey data from Mexico's National Water Quality Measurement Network (RENAMECA) and compare these results with population density data for municipalities within Morelos. I will also measure distances between towns and nearby water sources to observe if the most nearby source is clean or dirty based on RENAMECAs water sample data. With ArcGIS, I will



The state of Morelos on a map of Mexico (Source: Wikipedia)

use the analyze patterns toolbox to calculate density and the proximity toolbox to identify nearby water sources. This will allow me to assess spatial relationships and identify patterns of water accessibility across the state. Ultimately, this project aims to inform government and organizations such as Planet Water Foundation on which communities don't have access to clean water.

Methods & Approach

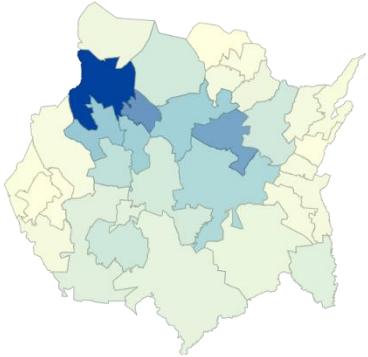
Using a quantitative research design, this project looks to analyze the relationship between clean accessible water and populations in the state of Morelos, Mexico. Primary data was collected from a Mexican government funded research group called RENAMECA, and population density data was collected by INEGI, a public body responsible for Mexico's 10-year census. To analyze the results of the datasets, a mixed use of observational and correlational research design was used to be able to both visually recognize trends in clean water locations and mathematically examine how the location of clean water correlates to nearby populations. Both designs will help investigate whether clean water accessibility in Morelos, Mexico, is correlated with population density. I hypothesize that more densely populated areas will have greater access to clean water compared to less populated regions.



CONAGUA gathering data from different types of water sources (Source: CONAGUA)

CONAGUA, a subsidiary of RENAMECA, conducts ongoing research and data collection on water quality throughout the entire country of Mexico. CONAGUA mainly studies major bodies of water by analyzing the physicochemical and microbiological parameters (CONAGUA). The presentation of the data consists of 4 subcategories of water: lotic, which includes rivers and streams, lentic, which includes lakes and reservoirs, coastal bodies, subterranean springs and wells. The entire dataset includes 7251 sites, 188 of which reside within the state of Morelos. Between 2012 and 2023, sites were sampled 127,425 unique times, gathering 446 different types of data. Water clarity and cleanliness specific results were individualized from the overall dataset, making analysis much easier and faster to do on computer programs.

Population data was sourced from the Mexican census of population and housing from 2020. The census itself consisted of 38 questions, which addressed multiple attributes such as characteristics of the housing units, characteristics of people, and classifiers such as religion, occupation, and many others. The study was conducted over 4 weeks, where over 147,000 people responded to the questionnaire (INEGI). The census was organized and carried out by the National System of Statistical and Geographical Information (SNIEG), which is regulated and coordinated by the National Institute of Statistics and Geography (INEGI). INEGI is Mexico's autonomous institute which collects and processes data on Mexico's territory, population, and economy (INEGI). Census results were separated out by State and municipality. Results from questionnaires also included detailed longitudinal and latitudinal coordinate information for ease of


Municipality populations for Morelos (Source: Data Mexico)

use when eventually mapping out the data points. Morelos itself has 1678 unique data entries, separating each municipality into urban areas. With such detailed data, extensive cleaning and sorting was conducted to only extract important information which was needed to calculate population density per town.

While the dataset that CONAGUA and RENAMECA provided was vast with many data points, it was difficult to decipher due to language barriers and unknown abbreviations. They offered an English version of the government website where the study was hosted, but the data file was not able to be translated into English. Luckily, I do know Spanish which eased the process greatly, but there was still difficulty with understanding niche academic and scientific vocabulary. Along with this, there were a ton of different results from surveys, and deciding which results to use was difficult. In a more advanced study, direct communication with CONAGUA and RENAMECA could have eased the data organization process, and they could have provided a translated version of their documents, along with a version that just included results from Morelos.

Preparation

Before initiating analysis on data sources, extensive research was conducted to discover and narrow down which datasets would be used. Beginning with clean water data, sources such as the United Nations had helpful statistics regarding sanitation and hygiene levels and provided helpful graphics of water data overtime. Along with this, the UN provided helpful distributions of water types within Mexico. Despite the extensive data sources and helpful visuals, they do not provide detailed enough data on unique municipalities and urban areas but instead look at the country of Mexico as a whole, and their data is not coordinated and prepared to be used within ArcGIS. Additionally, I discovered ongoing initiatives such as the Planet Water Foundation which deploy water filtration systems around Latin America and Asia. Hopefully this project can aid in their mission on bringing clean, accessible water to more people in Mexico by identifying where their services are most needed.

CONAGUAs data stood out as a valuable source due to its extreme detail and consistent results. As the project has been ongoing since 2012, they have vast experience and resources regarding sampling water sources. Furthermore, they include specific source coordinates and separate their results by both municipalities and urban areas, aiding in map projections later in the project. These coordinates can be paired with census, town boundary, and municipality boundary data to see which water source is most accessible for each town. To prepare this source for display and analysis, most of the data was omitted. Out of the 7251 sites in the dataset, 188 of them reside within Morelos. These sites were sampled from 1799 individual times ranging between 2012 and 2023. To maintain date consistency, only the most recent sample data was kept for each site. CONAGUA also offered 446 unique attributes for each sample. Out of the 446 attributes, 11 were chosen to decide whether water would be labeled as “clean” or “dirty”. The table below shows the attributes and what level each attribute needs to fall under, below, or in between to be labeled as “clean”.

Attributes	Alkalinity	Fecal coliforms	Escherichia coli	Oxygen Demand	Total Nitrogen	Water temperature
Clean	20-200 mg/L	Below 200 CFU/100mL	Below 50 mg/L	Below 0.5 mg/L	Below 10 mg/L	20-25°C

Attributes	Total Phosphorus	Surface salinity	Dissolved Oxygen %	Total Suspended Solids	Turbidity
Clean	Below 0.1 mg/L	Below 800 mg/L	Above 6 mg/L	Below 10 mg/L	Below 5 NTU

Attribute table with metrics representing clean water levels

When researching census data, Mexico's Secretary of Economy's website offered ample resources to find data which would fit the requirements of the project. It was also helpful that they offered their entire website in an English version, making navigation much simpler. One of their entities, Data Mexico, provided a system which allows users to view census data per state, and sources where the data originates from. This was incredibly helpful when analyzing their population and housing data, as all I had to do to access the raw dataset was go to the original source. SNIEG, which is a subsystem of INEGI, conducts a nation-wide census every 10 years. Within this census, they provide population information per town. SNIEG offers 1678 unique data entries, each relating to a town within Morelos. From SNIEG's main dataset, I extracted their municipalities shapefile, which included population count, and their localities shapefile, which also included population. All other data was omitted to clean up the analysis process.

With data sources collected and prepared, data can begin to be implemented into ArcGIS to be analyzed. Analysis will be conducted in 4 different ways resulting in 4 maps. The first map will just include water sample sites and display if they are clean or dirty. The second map will show population densities of municipalities and connect them to water sample locations, determining if more populated areas have cleaner water sources. The third map will connect localities with the closest water sample location, establishing if it is clean or dirty. The final map will connect localities and local water sources such as lakes or rivers. Using the water sample data, these water sources will be determined whether they are clean or dirty.

Potential challenges that might occur during data analysis might include gaps in the datasets. CONAGUAs dataset does not provide consistent attribute data, so multiple attributes are being used to fill in the gaps where they are detected. With the large amount of data supplied by both CONAGUA and INEGI, problems may occur during analysis by misinterpretation of results. To prevent this, the data was reviewed and organized with the help from an ArcGIS specialist at KU.

Conclusion

This proposed research aims to inform on water quality and access to said water. By using ArcGIS, this study will provide valuable insights on where institutions must focus their resources and efforts. In the future, this work can be expanded to other states to begin to equalize water disparity throughout the entire country of Mexico. In addition, primary data can be collected for more accurate and up-to-date information. This can include collecting a town's main source of water, average cost of getting water, and much more. By addressing the issue of water access in these communities, we can begin to create more accessible opportunities for vulnerable populations.

Why Morelos and this project? I chose the location and topic due to personal ties and experiences with the area. Growing up, I would visit my family in a town called Quebrantadero, which is located within Morelos. While commuting from the airport, wealth disparity was incredibly relevant, and even in the town, I could recognize how their access to water differed from how I accessed water back in the United States. Hopefully this research can be used to identify gaps in water infrastructure and organizations can begin the hard work of communicating with



Me at a young age in Quebrantadero

government and private systems. With more resources and data, we can help those in need, and make the world a more equitable, safer place.

If awarded the UGRA grant, my mentor and I will continue to stay in contact throughout the rest of the semester as I work on finishing organization of the data. Once organization is finished, the mapping process may begin and bi-weekly meetings will occur with my mentor, along with as-needed meetings, to prepare the maps. After map completion, a presentation will be prepared and presented at a relevant event. All other expectations will also be met, following a timeline prepared by me and my mentor. The project is estimated to be completed in early May.

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